Author: Imama Zubair AI Intern @ NETIXSOL

Week 4 Day 5

Adult Census Income Prediction

Executive Report

Business Goal

Predict whether an individual earns more than $50K a year using U.S. census data, in order to support targeted outreach. In this use case, reaching a smaller number of people who are more likely to qualify matters more than reaching everyone who might qualify, so Precision was prioritized over Recall throughout.

Data Used

Adult Census Income dataset (~45,000 records, 12 features used after dropping **education** and **fnlwgt**). Data was split 80/10/10 into train, dev, and test sets with stratification on the target. The target class is imbalanced, roughly 76% earning <=$50K and 24% earning >$50K.

Final Model & Deployment Artifact

The final model is a calibrated **LightGBM** classifier (isotonic calibration), chosen after comparing it against a tuned Random Forest, a Logistic Regression baseline, and a Stacking Ensemble (Random Forest + LightGBM, with Logistic Regression as the meta-model). The model is saved as **lgbm_calibrated_final_model.joblib** and wrapped in a **predict_income()** inference function that validates incoming data, applies the saved preprocessing pipeline, and returns a probability, predicted class, and the top-3 SHAP-driven features behind each individual prediction.

Key Hold-Out Metrics

- ROC AUC landed close together across all four models compared, between 0.88 and 0.90, meaning they all rank individuals similarly well.
- The calibrated LightGBM gave the best overall Precision/Recall balance at the chosen operating threshold.
- Class imbalance handling techniques (class weighting, random oversampling, SMOTE) were tested on top of LightGBM; all three raised Recall substantially, at some cost to Precision.

Chosen Threshold

A decision threshold of **0.60** was selected instead of the default 0.50, specifically to favor Precision, since false positives (outreach wasted on someone who doesn't actually qualify) are costlier than false negatives for this business use case.

Interpretability Findings

Global feature importance (both native LightGBM importance and permutation importance) consistently ranked **capital-gain** as the strongest single predictor, followed by **marital status** and **education-num** (an engineered feature combining education level and hours worked). SHAP confirmed the same ranking and added nuance: dots pushing predictions toward >$50K cluster around high capital-gain, being married, and higher education.

Local explanations on individual test cases showed that the model's one false positive examined was driven by the same features as its true positive (marital status and education-related signals looked identical), while its false negative had a weak or negative capital-gain signal that wasn't strong enough to flag an individual who actually earned >$50K. This suggests the model leans heavily on capital-gain as a shortcut, which can miss high earners whose income doesn't come through capital gains.

Fairness Observations

Selection rate (the share of a group predicted >$50K) differs meaningfully across protected groups. By sex, Male sits at 17.5% versus 4.6% for Female, nearly a 4x gap. By race, selection rate ranges from 6.1% (Black) up to 19.6% (Asian-Pac-Islander).

Precision stays reasonably consistent across groups (0.69 to 1.00), meaning the model is about equally reliable when it does predict >$50K. Recall is less even: Female (0.37) and Black (0.43) individuals who actually earn >$50K are flagged less often than other groups. This largely mirrors patterns already present in the training data rather than the model introducing new bias, but the gap is large enough that using raw predictions as-is for high-stakes decisions would carry forward that disparity.

Proposed mitigations:

- Do not use model output as a sole gate for high-stakes decisions; pair with human review for borderline cases.
- Consider group-specific threshold calibration to equalize recall across groups, accepting some precision trade-off.
- Track selection rate and recall by group on a recurring basis after deployment, not just once at launch.
- Treat the smallest groups (Other, Amer-Indian-Eskimo) with caution given limited sample sizes.

Recommended Next Steps

- Run an A/B test comparing model-flagged outreach leads (threshold 0.60) against the current outreach method over a 4-6 week window, measuring conversion rate and cost per qualified lead.
- Stand up the monitoring plan (data drift, prediction distribution, performance, fairness) before go-live.
- Revisit group-specific threshold calibration if the fairness gap is judged too large for this use case.
- Retrain on a 6-month cadence, or immediately if any monitoring alert threshold is breached and confirmed.

Conclusion

Overall, the calibrated LightGBM model met the project's goal by providing a strong balance between Precision and Recall while remaining interpretable and ready for deployment.